BUSINESS ANALYTICS PROJECT

Retail Sales, Customer and Profit Analytics

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1. Introduction

Business Analytics combines data, statistics, visualization and predictive methods to support business decisions. This project studies a realistic multi-channel retail business using a purpose-built synthetic transaction dataset.

2. Problem Statement

Management needs to understand sales and profit drivers, customer and regional patterns, the effect of discounts, future sales trends, operational risk and resource allocation.

3. Objectives

- Perform EDA and descriptive statistics.

- Measure relationships using correlation and regression.
- Quantify uncertainty using probability, sampling, confidence intervals and bootstrapping.
- Apply hypothesis testing, A/B testing and ANOVA.
- Forecast future sales using exponential smoothing and ARIMA.
- Solve a resource-allocation optimization problem.
- Use simulation and experimental design for business decisions.

4. Business Scenario / Domain Description

NovaMart Retail Analytics is a hypothetical multi-channel retailer selling Electronics, Home & Kitchen, Fashion, Beauty and Grocery through Website, Mobile App, Marketplace and Store channels. The project focuses on sales, profit, customer behavior, discounts, delivery and returns.

5. Dataset Description

The dataset contains 1,200 transaction records and 18 fields from 2024-01-01 to 2025-12-30. It was generated with fixed seed 42 and realistic relationships.

Field	Meaning	Type
Order_ID	Unique order	Categorical
Order_Date	Transaction date	Date
Customer_ID	Customer identifier	Categorical
Age	Customer age	Numerical
Region	Sales region	Categorical
Customer_Segment	Customer group	Categorical
Channel	Sales channel	Categorical
Payment_Method	Payment type	Categorical
Category	Product category	Categorical
Product	Product name	Categorical
Quantity	Units	Numerical
Unit_Price	Unit price	Numerical
Discount_Pct	Discount %	Numerical
Sales	Order value	Numerical
Profit	Estimated profit	Numerical
Rating	Customer rating	Numerical
Delivery_Days	Delivery time	Numerical
Returned	Return flag	Binary

6. Data Collection

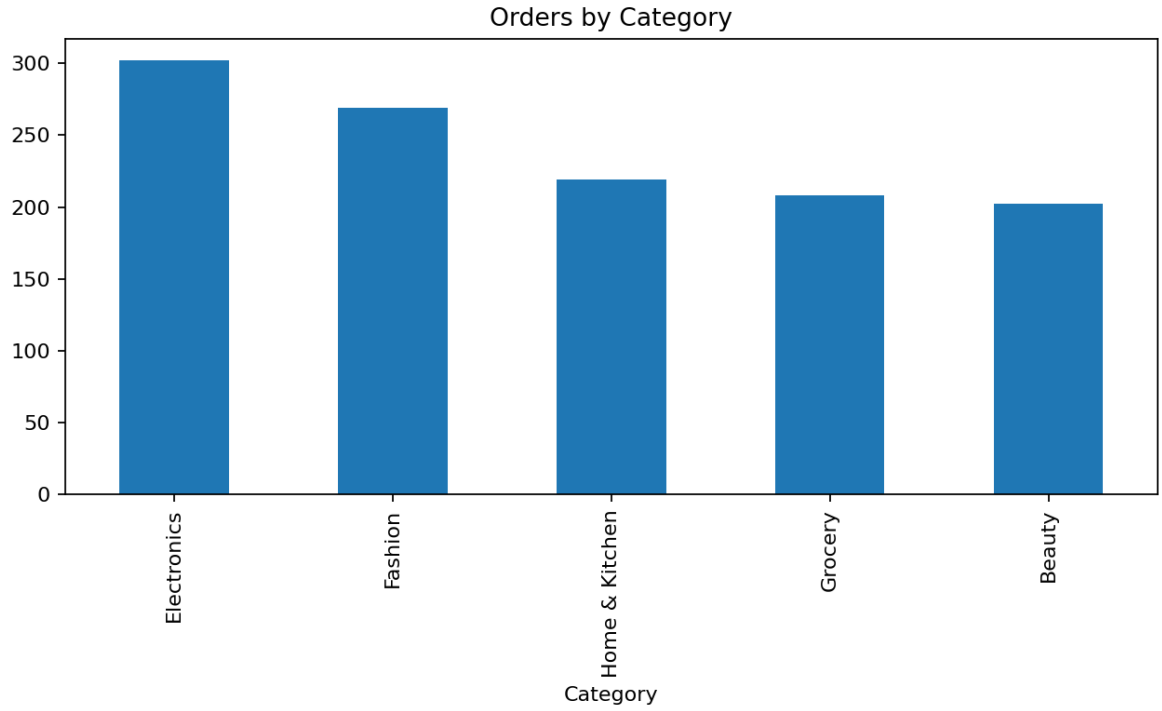
The dataset is synthetic and was generated specifically for this academic project. No personal or confidential information is used. Synthetic generation allows all required statistical methods to be demonstrated reproducibly.

7. Data Preprocessing

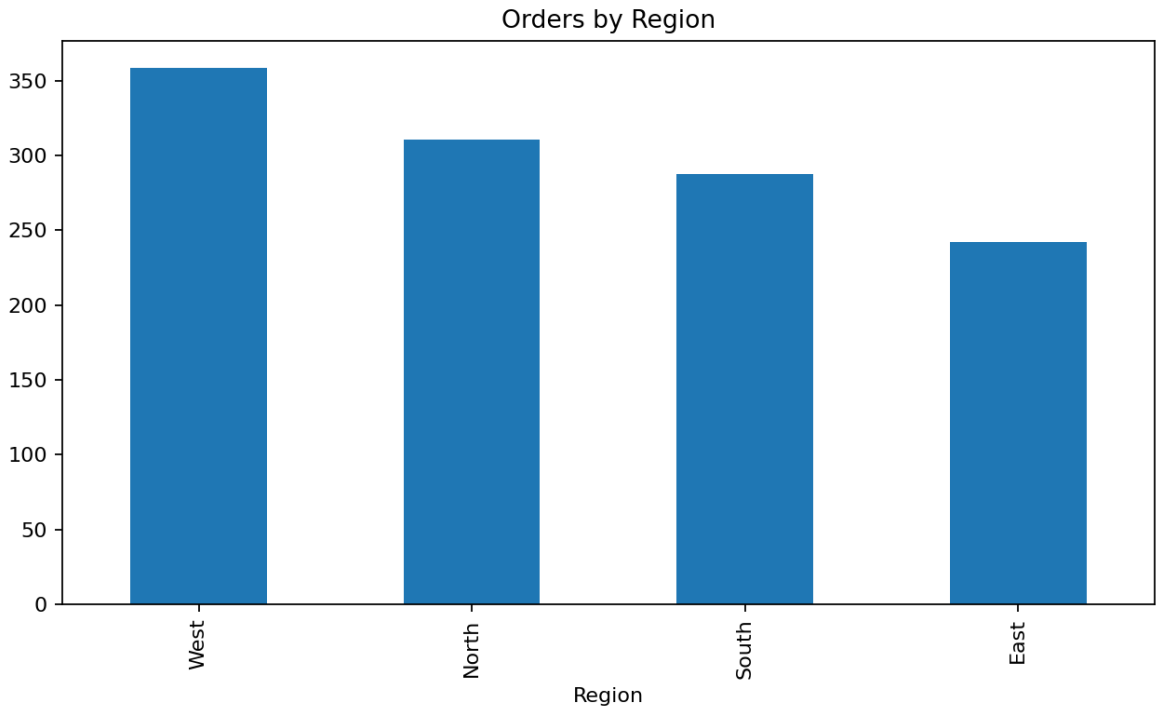
Missing values: 0. Duplicate rows: 0. Invalid sales values: 0. Invalid quantities: 0. Dates were converted to datetime and categorical/numerical fields were validated.

8. Exploratory Data Analysis

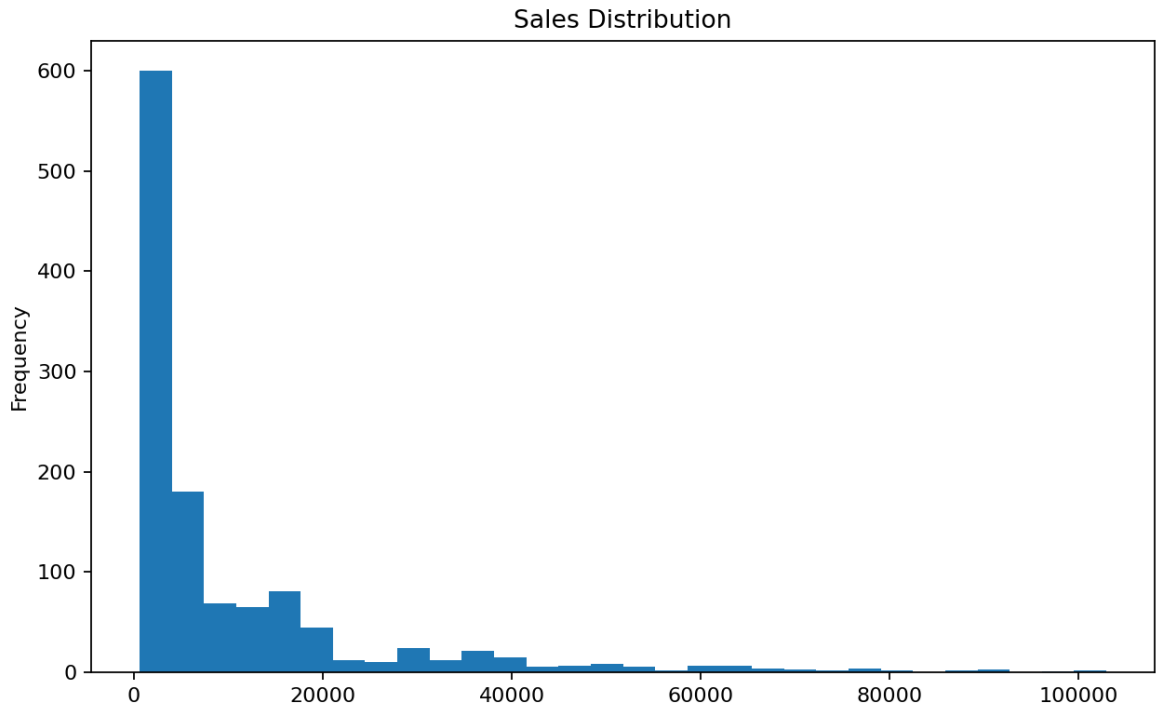
Measure	Value
Rows	1200
Columns	18
Categories	5
Regions	4
Channels	4
Segments	3



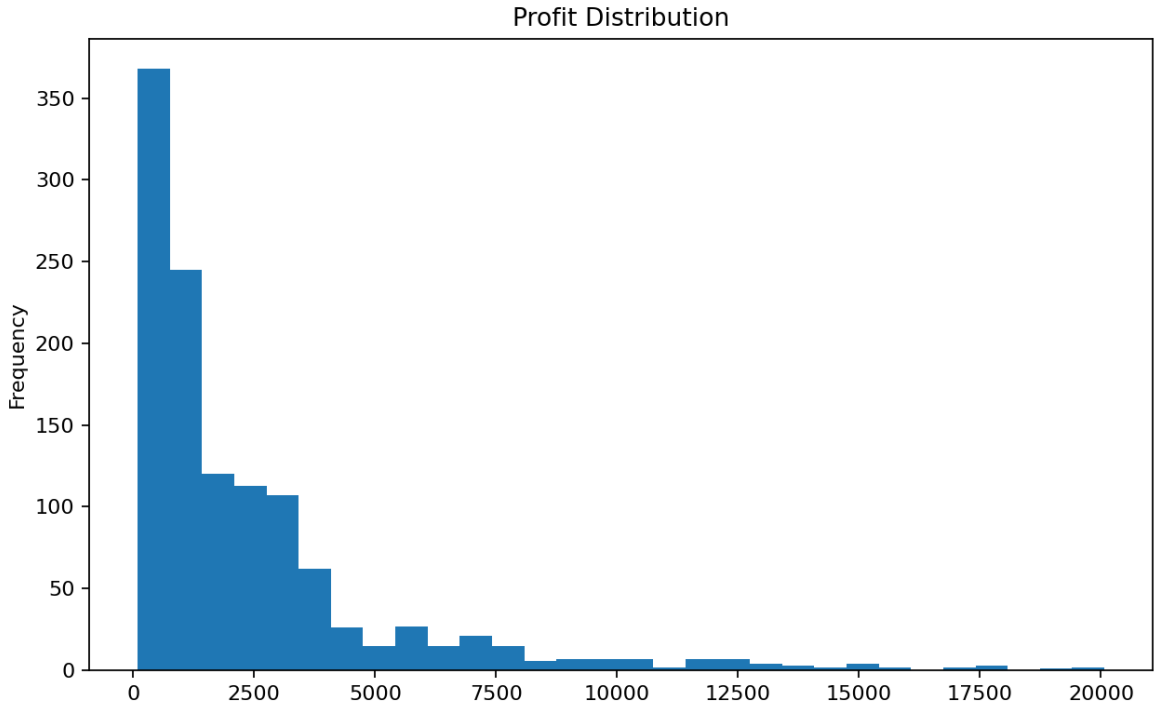
Observation: Category demand comparison.



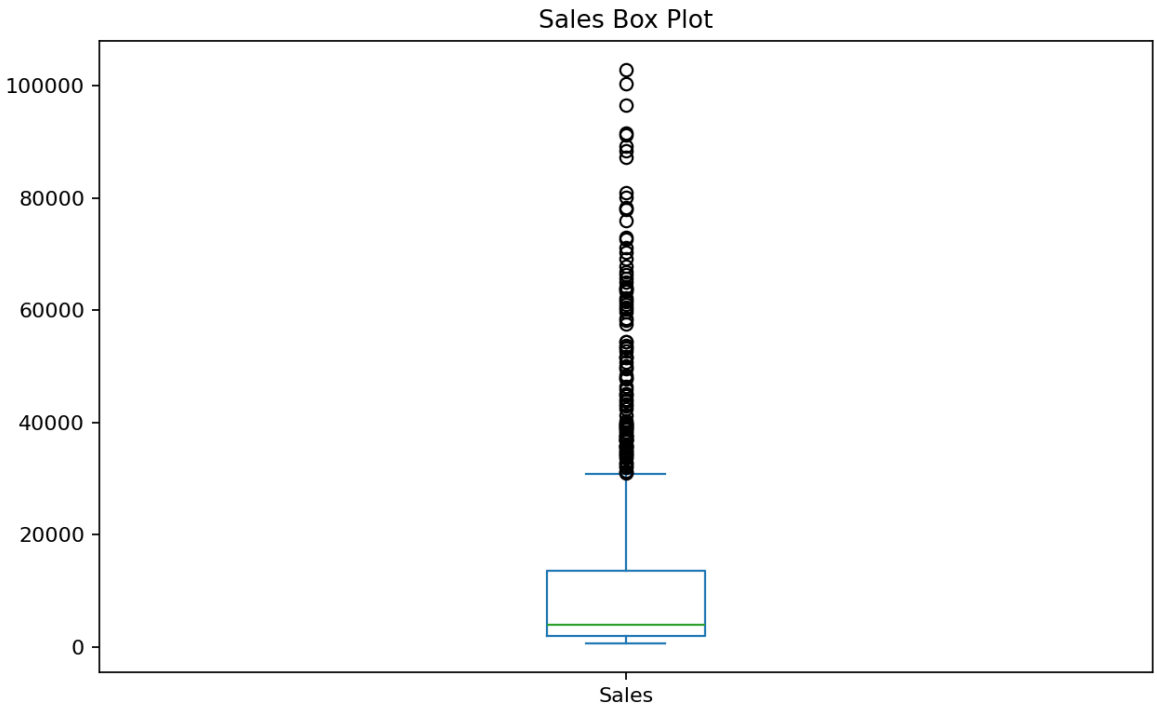
Observation: Regional order concentration.



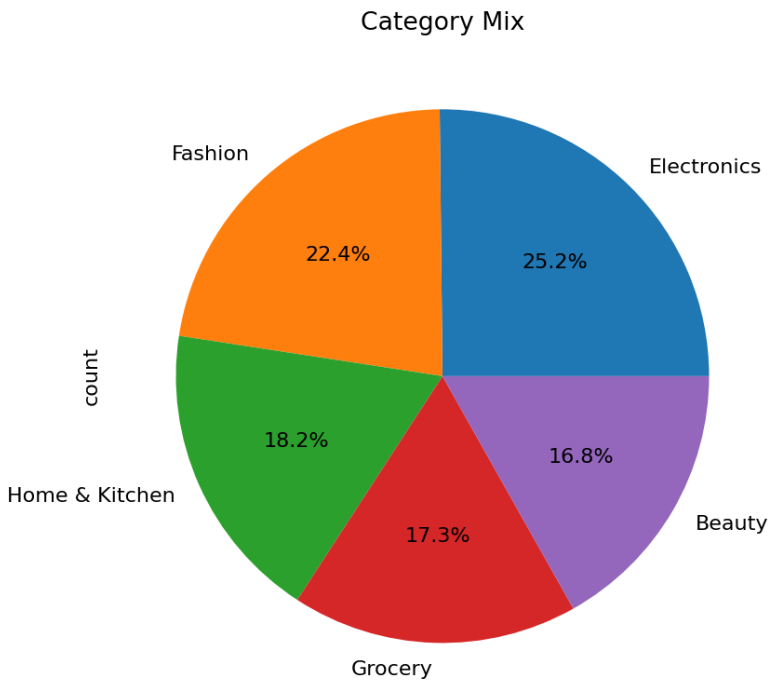
Observation: Sales are right-skewed with larger orders in the upper tail.



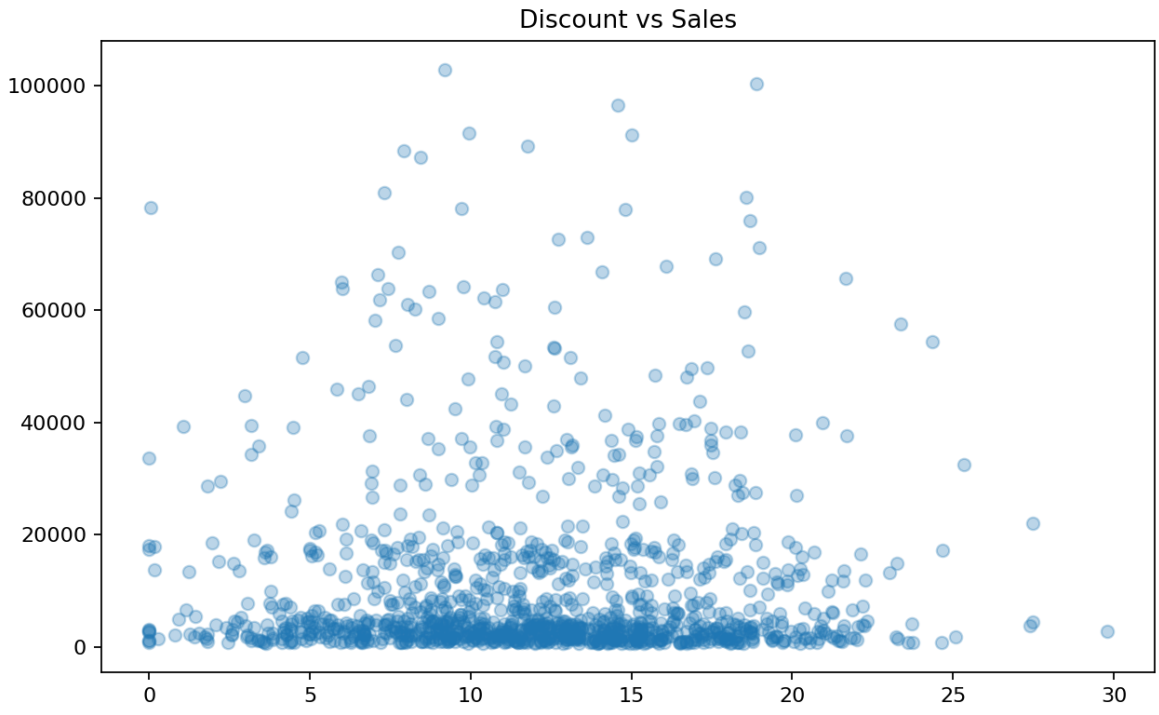
Observation: Profit variation across transactions.



Observation: Spread and high-value sales observations.



Observation: Relative category contribution.



Observation: Visual relationship between discounts and sales.

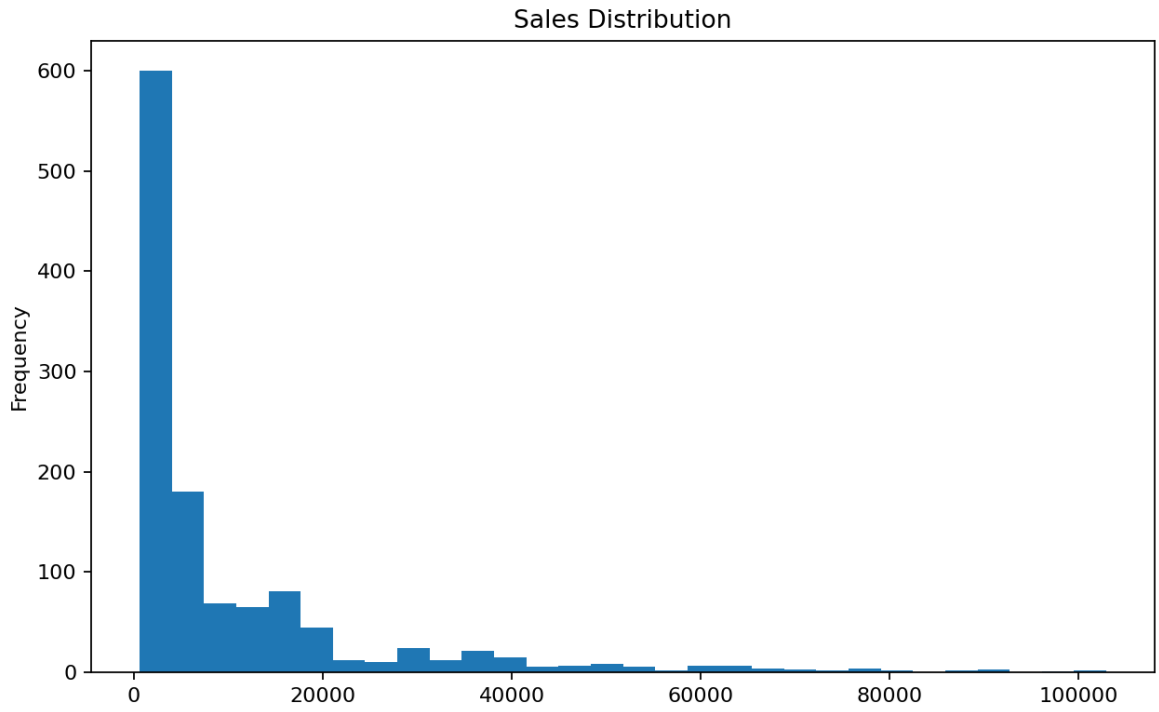
9. Descriptive Statistical Analysis

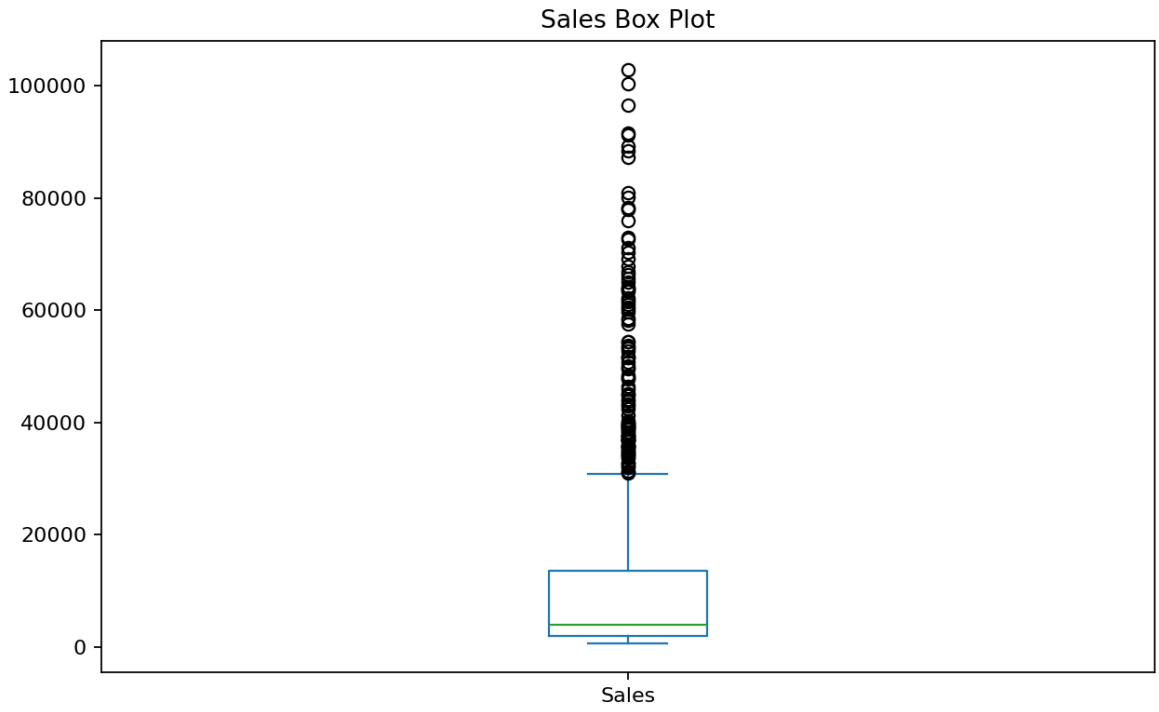
Variable	Mean	Median	Mode	Min	Max	Variance	Std Dev
Quantity	1.94	2.0	1	1	5	1.21	1.1

Unit_Price	6221.1	2434.33	808.78	706.26	22496.92	50728568.43	7122.4
Discount_Pct	12.05	12.08	0.0	0.0	29.78	25.98	5.1
Sales	10750.46	3995.16	626.11	586.83	102916.86	247078004.51	15718.72
Profit	2469.82	1390.72	97.05	90.96	20068.79	8997669.52	2999.61
Rating	4.09	4.1	4.0	2.8	5.0	0.2	0.45
Delivery_Days	4.47	4.0	5	1	10	2.87	1.69

Total sales = ₹12,900,549; total profit = ₹2,963,786; average order value = ₹10,750.46.

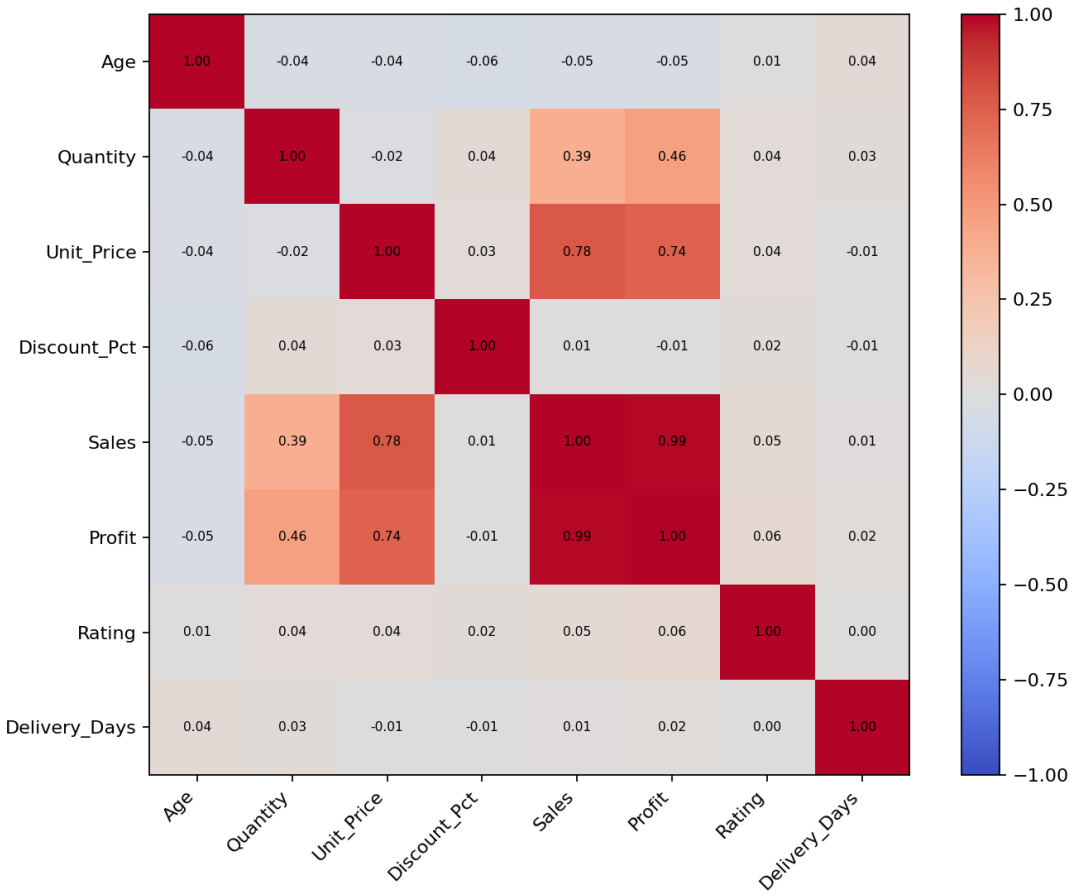
10. Distribution Analysis





Sales have a positive/right-skewed shape. The mean is above the median because a smaller number of high-value orders pull the average upward.

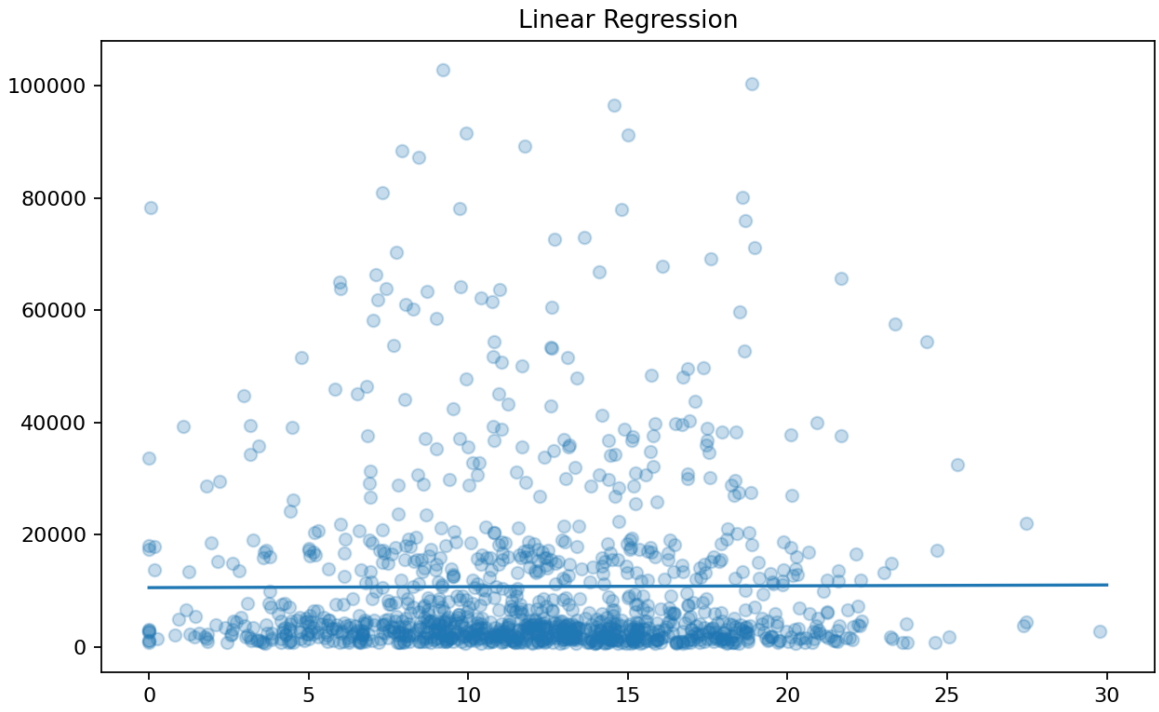
11. Correlation Analysis



Sales is strongly associated with Unit Price and Quantity because order value is constructed from these drivers. Profit also rises with sales, while discount is a weaker standalone explanatory factor.

12. Regression Analysis

Regression equation: Sales = 10,559.55 + (15.85 × Discount_Pct). R^2 = 0.000; RMSE = ₹15,711.96. The model estimates association, not causation.



13. Business Intelligence Dashboard

Recommended Tableau/Power BI dashboard: KPI cards for Sales, Profit, Orders, AOV and Return Rate; category bar chart; region/category pie chart; monthly sales line chart; and filters for Date, Region, Category, Channel and Customer Segment.

14. Probability and Probability Distributions

Probability that an order is above the 75th percentile of sales: 25.00%. Normal, uniform and exponential random distributions were also generated to demonstrate their characteristics.

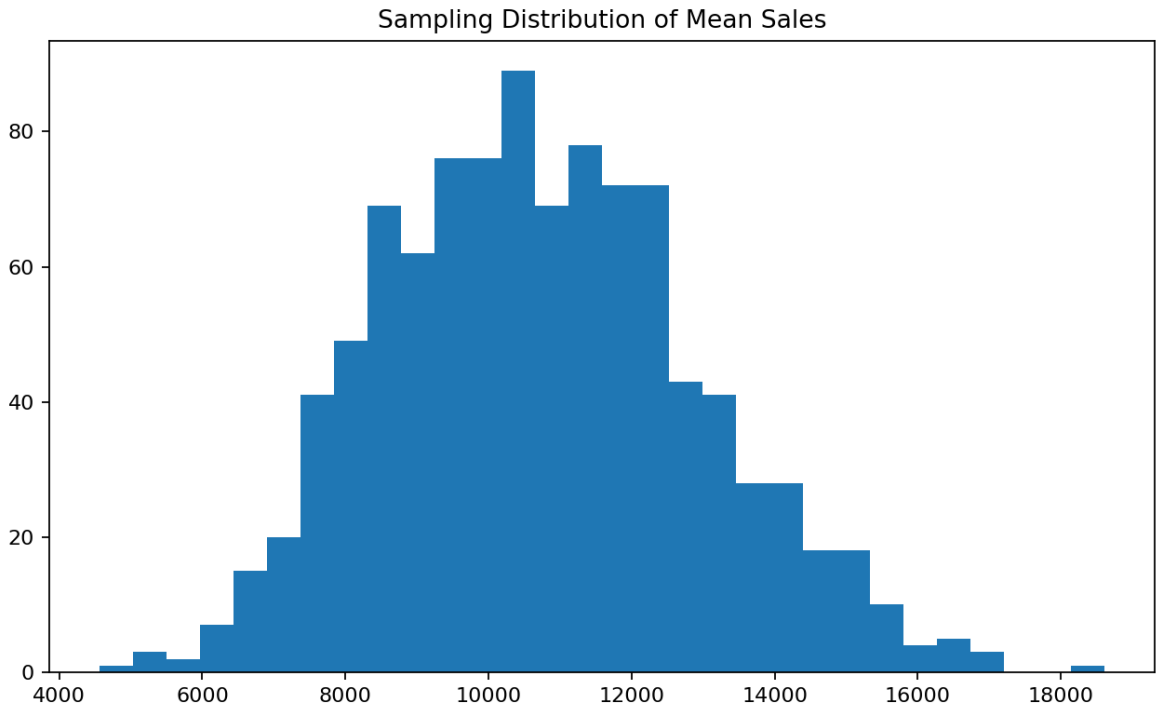
15. Decision Making Under Uncertainty

Alternative	High Demand	High Payoff	Low Demand	Low Payoff	Expected Value
Basic Campaign	60%	₹180,000	40%	₹60,000	₹132,000
Premium Campaign	60%	₹250,000	40%	₹20,000	₹158,000

Premium Campaign has the higher expected value, but management should consider its lower low-demand payoff.

16. Sampling and Sampling Distribution

From 1,000 samples of n=50, mean of sample means = ₹10,728.03; SD = ₹2,198.60; theoretical SE = ₹2,222.96. This illustrates the Central Limit Theorem.



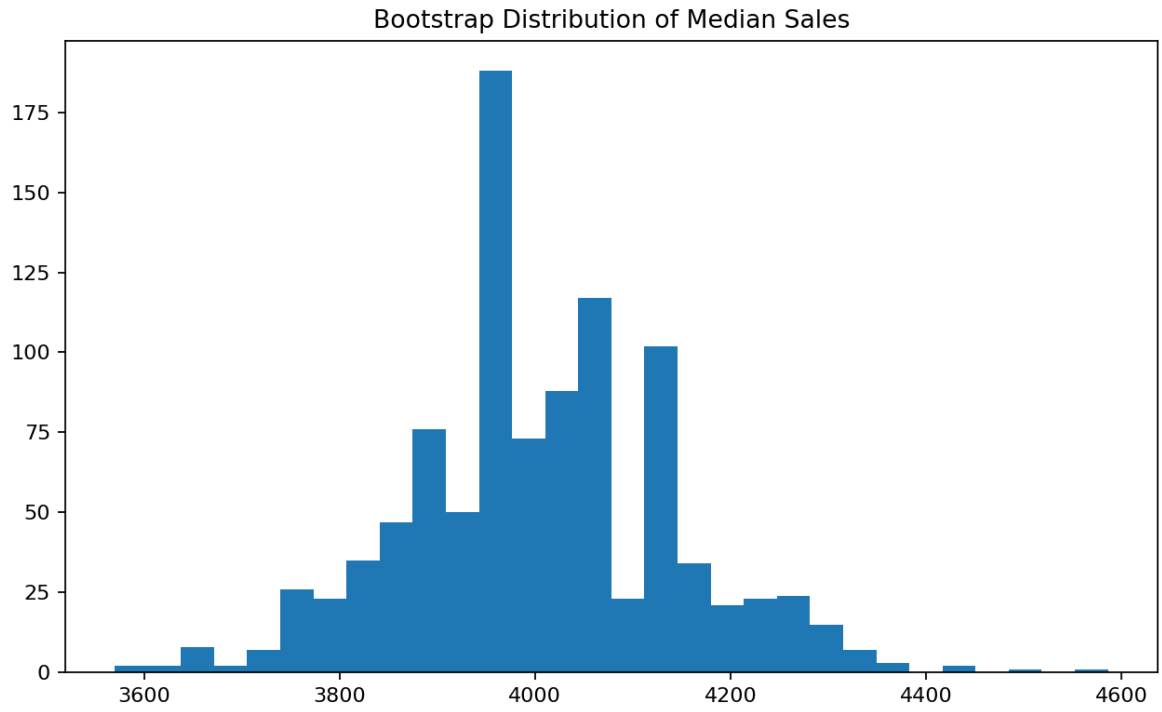
17. Confidence Interval Estimation

Level	Lower	Upper
90%	₹10,004.09	₹11,496.83
95%	₹9,861.10	₹11,639.81
99%	₹9,581.65	₹11,919.27

The 95% interval for mean order sales is approximately ₹9,861.10 to ₹11,639.81.

18. Bootstrapping

1,000 bootstrap resamples produced a 95% CI for median sales of ₹3,743.10 to ₹4,289.53.



19. Hypothesis Testing

Question: Is mean order sales different from ₹15,000? H_0: μ=₹15,000; H_1: μ≠₹15,000; α=.05.

One-sample t-test: t=-9.365, p=0.000000. Decision: Reject H_0.

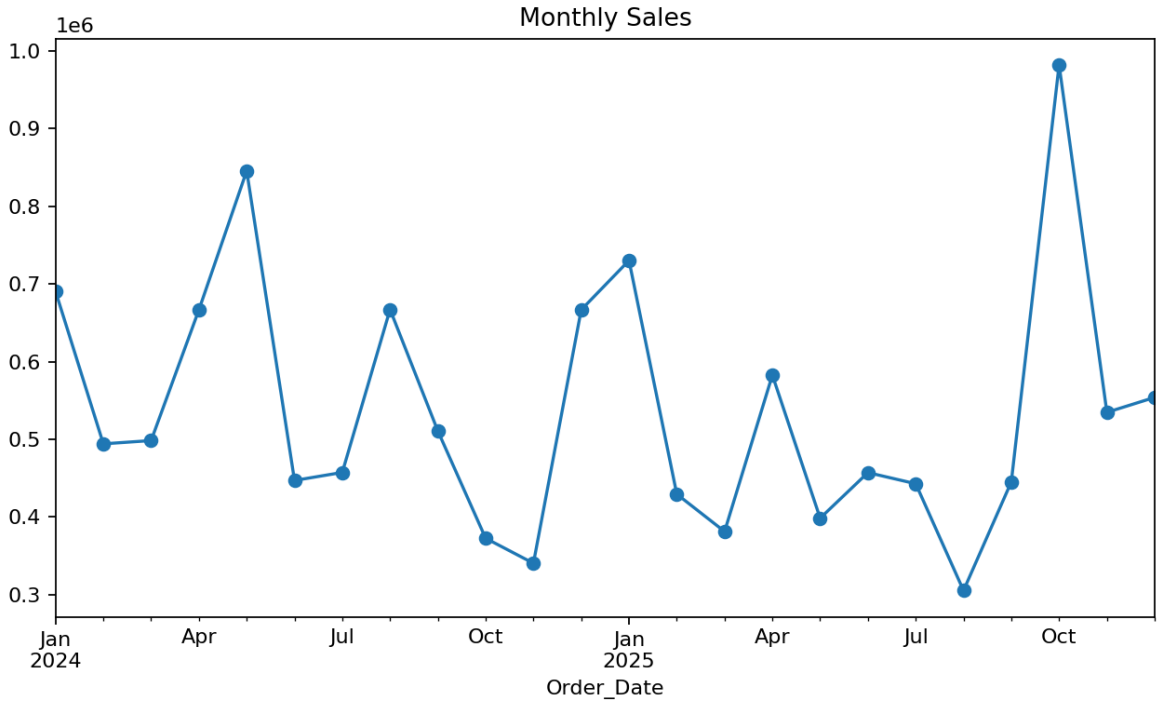
20. A/B Testing

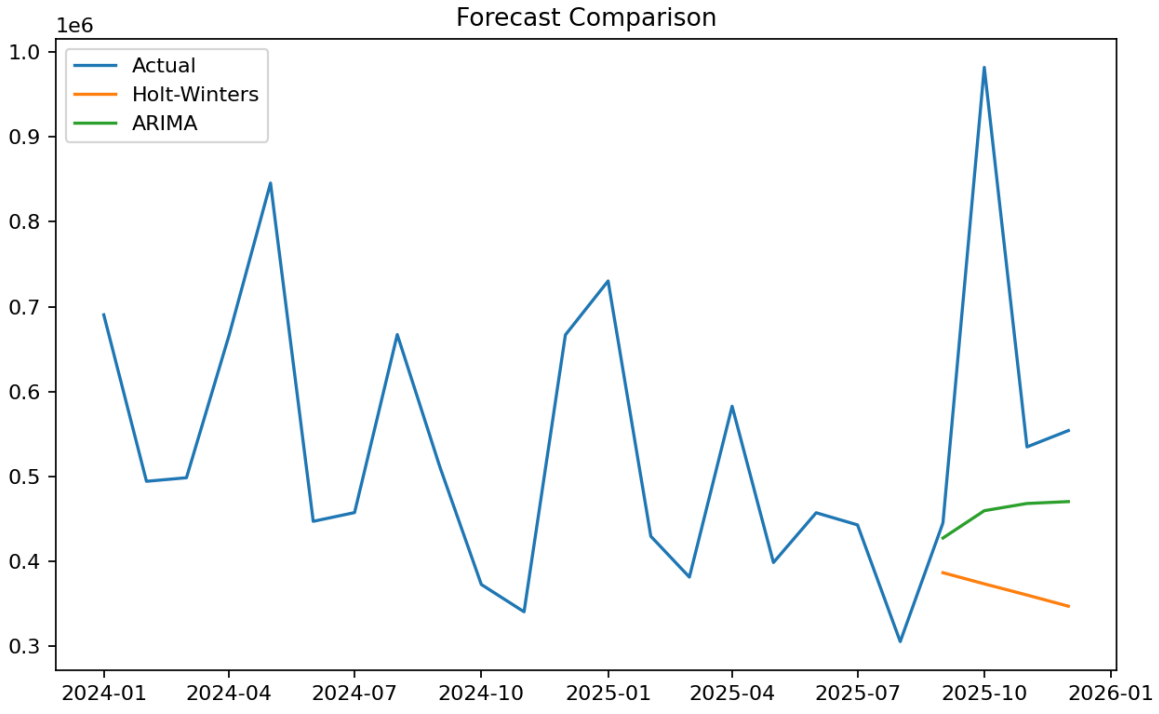
Because the sales dataset has no website impressions/conversions, this section uses a clearly labeled simulated controlled experiment: 1,000 visitors per variant.

Variant	Visitors	Conversions	Rate
A	1000	81	8.10%
B	1000	92	9.20%

Two-proportion z=-0.875, p=0.3816. The difference is not statistically significant at 5%.

21. Time Series Analysis





Monthly sales were analyzed for trend and forecasting. The final four months were held out as a test period.

22. Forecast Accuracy

Model	MAE	RMSE

Exponential Smoothing	₹262,137	₹334,271
ARIMA	₹172,620	₹266,723

Lower MAE and RMSE indicate better forecast accuracy on the holdout period.

23. Optimization Modeling

Maximize profit = 1800x + 1400y subject to 3x+2y≤180, 2x+4y≤160, x,y≥0. Optimal integer solution: x=50, y=15, maximum modeled profit = ₹111,000.

Business meaning: scarce labor and machine capacity should be allocated to the product mix that produces the highest feasible contribution.

24. Simulation Modeling

Stock Multiplier	Avg Shortage	Avg Units Served	Shortage Rate
1.0	0.17	29.29	0.56%
1.15	0.24	34.18	0.71%
1.3	0.23	39.15	0.59%

The simulation compares three inventory policies. Higher safety stock reduces shortage risk but may increase inventory holding cost.

25. ANOVA

One-way ANOVA across Website, Mobile App, Marketplace and Store: F=0.038, p=0.990098. Fail to reject H_o at α=.05; evidence is insufficient that channel means differ.

26. Experimental Design

A controlled experiment is proposed with two factors: Advertising Type (Social/Search) and Discount (5%/15%). The response variable is Sales and the experimental unit is a comparable campaign-market cell. Randomization and equal exposure should be used to reduce bias.

27. Factorial Experiment

Source	F-statistic	p-value
C(Advertising)	104.922	0.000000
C(Discount)	65.236	0.000000
C(Advertising):C(Discount)	1.158	0.284065

The factorial model estimates the main effect of advertising, the main effect of discount, and their interaction. These results are simulated experimental data and should not be presented as historical observations.

28. Business Insights and Findings

- Total modeled sales are ₹12,900,549.
- Total modeled profit is ₹2,963,786.
- Average order value is ₹10,750.46.
- Category and region distributions reveal where transaction volume is concentrated.
- Sales is strongly related to quantity and unit price, which are core basket-value drivers.
- The one-variable discount regression has R^2=0.000; discount alone does not explain all sales variation.

- The 95% confidence interval for mean sales is ₹9,861–₹11,640.
- The 95% bootstrap interval for median sales is ₹3,743–₹4,290.
- Forecast errors provide an objective basis for selecting the better forecasting method.
- Resource optimization produces a maximum modeled profit of ₹111,000 under the stated constraints.

29. Business Recommendations

- Monitor both sales and profit when evaluating categories and channels.
- Use monthly forecasts to plan inventory and staffing.
- Target promotions toward profitable customer, category and regional combinations.
- Do not assume larger discounts automatically create better outcomes; test them experimentally.
- Use dashboard filters for rapid investigation by date, region, category, channel and segment.
- Set safety stock using a target service level and expected demand variability.
- Repeat A/B testing using real website traffic before making permanent marketing changes.
- Use factorial experiments to identify the best combination of advertising and discount.

30. Conclusion

This project demonstrates a complete Business Analytics workflow using a purpose-built synthetic retail dataset. EDA, descriptive statistics, distribution analysis, correlation, regression, probability, decision analysis, sampling, confidence intervals, bootstrapping, hypothesis testing, A/B testing, time-series forecasting, forecast accuracy, optimization, simulation, ANOVA and experimental design were applied. The results demonstrate how descriptive, predictive and prescriptive analytics can support marketing, inventory, pricing, forecasting and resource-allocation decisions.

31. Appendix

Files included with the project package are the complete synthetic CSV dataset, generated charts, analysis summary and this Word report. Random seed 42 is used for reproducibility. No real customer personal information is contained in the dataset.

Suggested dashboard KPIs: Total Sales, Total Profit, Average Order Value, Orders, Return Rate and Average Rating. Suggested filters: Date, Region, Category, Channel and Customer Segment.